CHM111 Practice Exam 3 on chapter(s): 5, 6

1. 4.0% $\because$ $\Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
2. (lowest energy) red light < green light < violet light < x-rays (highest energy)
3. G → A $\because$ because this transition corresponds to the largest drop in energy levels, resulting in emission of a photon with the highest frequency (and shortest wavelength).
4. 44 kJ
5. 21 $\because$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.
6. 1s, 2p, 3s, 3p, 5s
7. 70 zJ $\because$ "B" is the level with the second-lowest energy, AKA the first excited state.
8. 6 $\because$ the number of lines appearing in the spectrum is equal to the number of transitions. Let n be the number of excited states; the number of transitions is $n(n+1)/2$.
9. 8.094×10^{-7} m
10. Yes $\because$ the de Broglie wavelength $\lambda = \frac{h}{mv}$ is larger than the width of the object.
11. 281 kJ
12. B → C $\because$ this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the longest wavelength and lowest frequency.
13. (lowest frequency) microwaves < infrared light < blue light < gamma rays (highest frequency)
14. D → C $\because$ because this transition corresponds to the smallest drop in energy levels, resulting in emission of a photon with the lowest frequency (and longest wavelength).
15. $1.\times 10^{3}$ zJ $\because$ "C" is the level with the third-lowest energy, AKA the second excited state.
16. -425.7 kJ. Since $\Delta H < 0$, the reaction is exothermic.
17. 3.89×10^{-7} m $\because$ $\Delta E = \left(\frac{-R_y}{n_2^2}\right) - \left(\frac{-R_y}{n_1^2}\right)$ and $\Delta E = \frac{hc}{\lambda}$
18. -3.93×10^{4} J
19. Absorbed $\because$ a photon is absorbed when an electron transitions to a higher energy level
20. (shortest wavelength) x-rays < green light < yellow light < microwaves (longest wavelength)
21. C → B $\because$ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
22. 2p: $n = 2$, $l = 1$, $N_e = 6$ ... 4d: $n = 4$, $l = 2$, $N_e = 10$ ... 3p: $n = 3$, $l = 1$, $N_e = 6$
23. 30 zJ $\because$ "A" is the level with the lowest energy, AKA the ground state.
24. A → G $\because$ this transition corresponds to the largest energy difference, resulting in absorption of a photon with the shortest wavelength and highest frequency.

25. B → C ∵ because this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the lowest frequency (and longest wavelength).

26. 3.053 m

27. A → D ∵ because this transition corresponds to the largest energy difference, resulting in absorption of a photon with the highest frequency (and shortest wavelength).

28. G → A ∵ this transition corresponds to the largest energy difference, resulting in emission of a photon with the shortest wavelength and highest frequency.

29. 9.93×10^{-7} m ∵ the energy of the emitted photon is equal to the difference in energy levels and $E = hc/\lambda$

30. 80 zJ ∵ the energy of the emitted photon is equal to the difference in energy levels. $200 - 120 = 80$

31. 0.13 J/g/K

32. 11 J